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NPTEL (https://swayam.gov.in/explorer?ncCode=NPTEL) » The Joy of Computing using Python (course)

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Course outline

About NPTEL ()

How does an NPTEL online course work? ()

Week 1 ()

Week 2 ()

Week 3 ()

week 4 ()

Week 5 : Assignment 5

The due date for submitting this assignment has passed.

Due on 2025-02-26, 23:59 IST.

Assignment submitted on 2025-02-26, 04:58 IST

1) Which of the following is the correct way to add data with key as $\mathbf{CS102}$ and value as $\mathbf{"Database\ Management\ Systems"}$ to a dictionary named $\mathbf{courses}$? *1 point*

- $\mathbf{courses["CS101"] = ("Database\ Management\ Systems")}$
- $\mathbf{courses["CS102"]["Database\ Management\ Systems"]}$
- $\mathbf{courses["CS102"] = "Database\ Management\ Systems"}$
- $\mathbf{courses["CS101"] = "Database\ Management\ Systems"}$

Yes, the answer is correct.
Score: 1

Accepted Answers:
$\mathbf{courses["CS102"] = "Database\ Management\ Systems"}$

2) What is the probability of Monty not opening the door with goat, given the hypothesis that you initially chose the door which has car ? *1 point*

- 0
- 0.5
- 1
- 0.33

Week 5 ()

- Introduction to Dictionaries (unit?unit=104&lesson=105)
- Speech to Text : No need to write 01 (unit?unit=104&lesson=106)
- Speech to Text : No need to write 02 (unit?unit=104&lesson=107)
- Speech to Text : No need to write 03 (unit?unit=104&lesson=108)
- Monte Hall : 3 doors and a twist 01 (unit?unit=104&lesson=109)
- Monte Hall : 3 doors and a twist 02 (unit?unit=104&lesson=110)
- Rock, Paper and Scissor : Cheating not allowed !! 01 (unit?unit=104&lesson=111)
- Rock, Paper and Scissor : Cheating not allowed !! 02 (unit?unit=104&lesson=112)

Yes, the answer is correct.
Score: 1

Accepted Answers:
0

3) What should be replaced with `?` in line 10, so that there is high chance that **final_choice** is equal to 2 ? ***1 point***

```
import random
initial_choice = random.randint(0, 2)
doors = ['goat', 'goat', 'car']

for i in range(3):
    if i != initial_choice and doors[i] != 'car':
        monty_opens = i
        break

if ? :
    for i in range(3):
        if i != initial_choice and i != monty_opens:
            final_choice = i
            break
else:
    final_choice = initial_choice
```

- [x] 2
- [ ] $2 * *(1024)$ % 2 != 0
- [x] $2 * (2 * *(89) + 7)$ % 2 == 0
- [ ] $(2 * *(90))$ % len(doors) != 1

Yes, the answer is correct.
Score: 1

Accepted Answers:
2
$2 * (2 * *(89) + 7)$ *% 2 == 0*

4) Given that you have a sorted list of 2048 elemets,what is the maximum number of comparisons to search such an element using linear search ? Can binary search do it in less number of comparisons ? ***1 point***

- ( ) Yes, binary search can do it more efficiently, linear search takes 1024 comparisons
- (•) Yes, binary search can do it more efficiently, linear search takes 2048 comparisons
- ( ) No, binary search may/may not do it more efficiently, linear search takes 1024 comparisons
- ( ) No, binary search will take more comparisons, linear search takes 2048 comparisons

Yes, the answer is correct.
Score: 1

Accepted Answers:
Yes, binary search can do it more efficiently, linear search takes 2048 comparisons

Rock, Paper and Scissor : Cheating not allowed !! 03 (unit?unit=104&lesson=113)

Rock, Paper and Scissor : Cheating not allowed !! 04 (unit?unit=104&lesson=114)

Sorting and Searching : 20 questions game 01 (unit?unit=104&lesson=115)

Sorting and Searching : 20 questions game 02 (unit?unit=104&lesson=116)

Sorting and Searching : 20 questions game 03 (unit?unit=104&lesson=117)

Sorting and Searching : 20 questions game 04 (unit?unit=104&lesson=118)

Sorting and Searching : 20 questions game 05 (unit?unit=104&lesson=119)

Sorting and Searching : 20 questions game 06 (unit?

5) What is the primary use of a file with a .wav or .wave extension? ***1 point***

- Storing video recordings
- Storing log files
- Storing waves of files
- Storing audio recordings

Yes, the answer is correct.
Score: 1

Accepted Answers:
Storing audio recordings

6) What does this program print in the end ? ***1 point***

```
import random
n = 10
counter = 0
for i in range(n):
  choices = ["rock", "paper", "scissors"]
  choice1 = random.choice(choices)
  choice2 = random.choice(choices)
  if choice1 != choice2:
    counter += 1
print(counter/n)
```

- Fraction of throws where both players showed different symbol(rock/paper/scissors)
- Fraction of throws where both players showed rock.
- Fraction of throws where both players showed same symbol(rock/paper/scissors)
- Fraction of throws where both players showed paper.

Yes, the answer is correct.
Score: 1

Accepted Answers:
Fraction of throws where both players showed different symbol(rock/paper/scissors)

7) In binary search the list gets divided in every iteration to find the element. ***1 point***

- True
- False

Yes, the answer is correct.
Score: 1

Accepted Answers:
True

8) Which of the following are requirements for binary search to work correctly? ***1 point***

- The list can be unsorted.
- The list must contain only positive integers.

unit=104&lesson=120)

Sorting and Searching : 20 questions game 07 (unit?unit=104&lesson=121)

Sorting and Searching : 20 questions game 08 (unit?unit=104&lesson=122)

Week 5 Feedback Form: The Joy of Computing using Python (unit?unit=104&lesson=123)

Quiz: Week 5 : Assignment 5 (assessment?name=519)

Week 5: Programming Assignment 1 (/noc25_cs69/progassignment?name=520)

Week 5: Programming Assignment 2 (/noc25_cs69/progassignment?name=521)

Week 5: Programming Assignment 3 (/noc25_cs69/progassignment?name=522)

Week 6 ()

- The list must be sorted.
- The list must have no duplicate values.

Yes, the answer is correct.
Score: 1

Accepted Answers:
The list must be sorted.

9) Given an array [4, 2, 7, 1, 3], what will be the array after the third pass of Bubble Sort? ***1 point***

- $[1, 3, 2, 4, 7]$
- $[2, 4, 1, 3, 7]$
- $[1, 2, 3, 4, 7]$
- $[2, 1, 3, 4, 7]$

Yes, the answer is correct.
Score: 1

Accepted Answers:
$[1, 2, 3, 4, 7]$

10) How many swaps are performed in iteration 3 for bubble sorting list [5,4,3,2,1]? Enter 0 if the list is sorted in less than 3 iterations.

2

Yes, the answer is correct.
Score: 1

Accepted Answers:
(Type: Numeric) 2

1 point

Week 7 ()

Week 8 ()

Week 9 ()

Week 10 ()

Week 11 ()

Week 12 ()

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